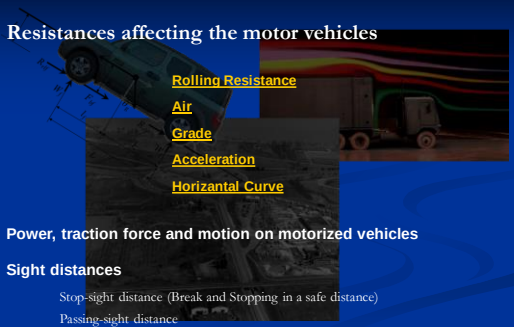
VEHICLE MOVEMENTS
Weekly Subjects and Related Preparation Studies

WEEK	SUBJECT	DATE	
1	INTRODUCTION BASIC DEFINITIONS (TERMS)	05.10.2020 08.10.2020	Monday Thursday
2	ROAD USER CHARECTERISTICS VEHICLE MOVEMENTS	12.10.2020 15.10.2020	Monday Thursday
3	TRAFFIC FLOW AND CAPACITY ROADWAY CLASSIFICATION AND GEOMETRIC STANDARDS	19.10.2020 22.10.2020	Monday Thursday
4	HIGHWAY ROUTE STUDY **NATIONAL HOLIDAY**	26.10.2020 29.10.2020	Monday Thursday
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution) HORIZANTAL CURVE AND SUPERELEVATION	02.11.2020 05.11.2020	Monday Thursday
6	**APPLICATION (VEHICLE MOVEMENTS)** **ASSIGNMENT - APPLICATION (TRAFFIC)**	09.11.2020 12.11.2020	Monday Thursday
7	**APPLICATION (HORIZANTAL CURVE)** **ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)**	16.11.2020 19.11.2020	Monday Thursday
8	**MID TERM EXAM**	23.11.2020 26.11.2020	Monday Thursday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN) TRANSITION CURVES– ASSIGNMENT (RETURN)	30.11.2020 03.12.2020	Monday Thursday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE VERTICAL ALIGNMENT AND VERTICAL CURVE	07.12.2020 10.12.2020	Monday Thursday
11	**APPLICATION (TRANSITION CURVES)** **ASSIGNMENT**	14.12.2020 17.12.2020	Monday Thursday
12	**APPLICATION (VERTICAL CURVE)** **APPLICATION (VERTICAL AND HORIZANTAL CURVE)**	21.12.2020 24.12.2020	Monday Thursday
13	PLANNING OF INTERSECTION AND URBAN ROADS PROJECTS DELIVERY DUE DATE (December 30, 2020)	28.12.2020 31.12.2020	Monday Thursday
14	DRAINAGE IN HIGHWAY **MAKEUP EXAM**	04.01.2020 07.01.2020	Monday Thursday

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles



- Rolling Resistance
- Air
- Grade
- Acceleration
- Horizantal Curve

Power, traction force and motion on motorized vehicles

Sight distances

Stop-sight distance (Break and Stopping in a safe distance)

Passing-sight distance

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

Rolling (trundling) (Dy) (kg)

The main reason of the rolling resistance is the friction between the vehicle's power trains, tyres contact points with pavement and the air circulation around the tyres.

$$Dy = \mu y * Q$$

Q- Weight of the vehicl (kg)

μy – Rolling resistance (kg/kg)

μy = f (speed, wearing course type, road surface condition etc.)

If the speed of the vehicle is below 50 km/h, rolling resistance coefficient can directly be taken from the table below.

(μo); Beton Yol 0,01 – 0,02

Asfalt 0,02 - 0,025

Stabilize 0,03 - 0,05

Parke 0,04 - 0,05

V > 50km/h **μy= (1-(0,01*(V-50)))* μo**

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

Air resistance (Dh)

If any vehicle moves below effects occures around the vehicle;

- The air pressure affecting perpendicular to the vehicle's cross-section in the opposite direction of the vehicle movement
- Friction resulting from the turbulenced air around the vehicle
- The air generated by the radiator and ventilator on the vehicle

$$Dh = \frac{\rho * C * F * V^2}{13}$$

$$\rho = \frac{1,225\ kg/m^3}{9,81\ m/sn^2}$$

$$F = 0,90 * b * h$$

Turbulence around the body (85%)
Air friction (12%)
Air flow through components (3%)

C =A coefficient related to the vehicle's shape and surface smoothness
ρ= The specific mass of the air at the sea level and 15 °C(kg sn² / m⁴)
V= Speed of the vehicle (m/sn)
F = Aracın kesit alanı (m²)
b: Width of the vehicle (m)
h: Height of the vehicle (m)

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

$$Dh = \frac{\rho * C * F * V^2}{13}$$

$$Dh = \frac{0,125}{2} * C * F * \frac{V^2}{3,6^2} = 0,0048 * C * F * V^2$$

$$k = \rho * C$$

$$Dh = \frac{k * F * V^2}{12,96}$$

$$Dh(kg) = 0,0048 * C * F(m^2) * V^2(km/sa)$$

If the unit of the air resistance is taken as Newtons, the formula becomes:

$$Dh(N) = 0,75 * C * F * V^2$$

VEHICLE MOVEMENTS

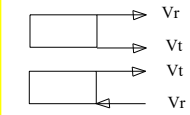
Limits of C and F values for various types of vehicles are given on the table below.

Vehicle type	C (kg.sec²/m⁴)	F (m²)
Racing cars	0,010-0,015	1,5-2,0
Normal automobiles	0,015-0,030	1,5-2,6
Busses	0,025-0,050	4,0-6,5
Trucks	0,050-0,070	3,0-6,0

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

Effcets of the wind



$$Dh = \frac{\rho}{2} * C * F * (Vt - Vr)^2$$

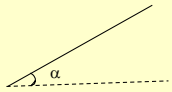
$$Dh = \frac{\rho}{2} * C * F * (Vt + Vr)^2$$

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

Grade Resistance (De)

Grade Resistance (De)



$De = Q*\sin\alpha \cong Q*\tan\alpha$

$\tan\alpha = S$ = Grade of the roadway

De = (+) Q*S (upgrade)
(-) (downgrade)

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

Acceleration Resistance (Di)

$$D_i = \frac{Q}{g} * \gamma$$

γ: Acceleration of the vehicle (m/sn^2)

Q (kg) g=9,81 m/sn^2

VEHICLE MOVEMENTS

Resistances affecting the motor vehicles

Horizantal Curve Resistance (Dk)

While moving on horizontal curves, the frontal wheels are rotated with an angle to the axis. The curve resistance occurs on the contact point of the frontal wheels with the pavement surface. Curve resistance increases wth the curvature and the speed of the vehicle.

R (m)	V (km/sa)	Dk (kg)
350	80	18
350	96	36
175	48	18
175	64	54

VEHICLE MOVEMENTS

Rolling Resistance

Dy= μy *Q

Air

$$Dh(kg) = 0{,}0048 * C * F(m^2) * V^2(km/sa)$$

Grade

De = (+) Q*S (upgrade)
(-) (downgrade)

Acceleration

$$D_i = \frac{Q}{g} * \gamma$$

γ: Acceleration of the vehicle (m/sn^2)

Q (kg) g=9,81 m/sn^2

Horizantal Curve

R (m)	V (km/sa)	Dk (kg)
350	80	18
350	96	36
175	48	18
175	64	54

VEHICLE MOVEMENTS

Power, traction force and motion on motorized vehicles

VEHICLE MOVEMENTS

Power, traction force and motion on motorized vehicles

If the traction force is Z, speed is V and, power is N for a vehicle,

N = Z*V Z→ kg V → m/sn V (km/h) / 3,6 = (m/sn)

$75 * N_B = Z * \frac{V}{3,6}$ $N_B = \frac{Z * V}{270}$ $Z = \frac{270 * N_B}{V} * \eta$

η→efficiency coefficient (< 1)

To start the movement, the resistances should be overtaken by the traction force. This is named as the first condition of the motion and can be formulated as;

Z ≥ Σ D

Z > Dy + Dh + De + Dk

VEHICLE MOVEMENTS

Power, traction force and motion on motorized vehicles

To start the movement, the resistances should be overtaken by **the traction force**. This is named as the first condition of the motion and can be formulated as;

Z ≥ Σ D

Z > Dy + Dh + De + Dk

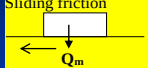
If the vehicle is moving with a constant speed on a curveless and graded section, the formula can be written as : Z ≥ Dy + Dh +De
If the vehicle moves with a constant speed on slopeless and straight section, the formula can be written as : Z ≥ Dy + Dh
If the vehicle to start a stopping vehicle moves on slopeless and straight section in , the formula can be written as : Z ≥ Dy

VEHICLE MOVEMENTS

To start a stopping vehicle to move

First Condition Z ≥ΣD

Sliding friction



Sliding friction force = Q_m*f

Q_m: The adherance weight or the weight of the vehicle that is carried by the motor wheels
f : Sliding friction coefficient

The wheels that the engine drives.

Car Qm ≅ 1/2 * Q
Heavy Vehicle Qm ≅ 2/3 * Q

First condition, to make a vehicle in a motion Z > Σ D

Second condition, to make a vehicle in a motion Qm*f ≥ ∑ D

VEHICLE MOVEMENTS

According to the two conditions of motion, for a stopping vehicle the following situations can be written:

Z, ΣD, Qm*f

1) ΣD < Z < Qm*f	normal motion
2) ΣD < Qm*f < Z	Forward motion with spinning
3) Qm* f < ΣD < Z	Spinning without motion
4) Z < ΣD < Qm*f	No motion

VEHICLE MOVEMENTS

Sliding friction coefficient depend on road surface

Surface Type	Kuru	Islak Temiz	Islak Hafif Çamurlu
Asphalt	0,60	0,35	0,25
Concrete	0,75	0,55	0,40
Paving stone	0,60	0,40	0,30
Basalt	0,45	0,25	0,15
Granit	0,50	0,35	0,25
Stabilized	0,70	0,50	0,40

Sliding friction coefficient in winter

	Dried		Wet	
	w/o Chain	w Chain	w/o Chain	w Chain
Snow	0,45	0,55	0,10	0,20
İce	0,15	0,35	0,05	0,15

VEHICLE MOVEMENTS

Question

A truck is moving along a 8% upgraded road section with a constant speed of 40 km/h. The motor power of the truck is 170kW and the weight of the truck is 5 tons.

A) Find the traction force Ztr.
B) Find the amount of useful load that the truck can carry.
C) If this truck stops on the road, can it move again with the useful load calculated in part

Given: rolling resistance coefficient, μ_r=0,030; air resistance coefficient, K=0,070; F=0,9.b.h; b=2,30 m, h=2,70 m;

motor efficiency coefficient, η=0,85, Sliding friction coefficient, *f*=0,60; The adherence weight, Q_m=2/3.Q.

$$W_h = 0{,}75.K.F.V^2$$

VEHICLE MOVEMENTS

a) $Ztr = 3600 * \left(\frac{Nm}{V}\right) * \eta$ ➔➔ $Ztr = 3600 * \left(\frac{170}{40}\right) * 0{,}85$ ➔➔ $Ztr = 13005\ N$

b)

$Ztr = \sum W\ ;\ \sum W = Wy + Wh + We$

$Wy = \mu_r * Q$ ➔➔ $0{,}030 * Q$ ➔➔ $0{,}03Q$

$W_h = 0{,}75.K.F.V^2$ ➔➔ $0{,}75*0{,}07*(0{,}9*2{,}30*2{,}70)*40^2$ ➔➔ $469{,}48\ N$

$We = \left(\frac{Q*s}{100}\right)$ ➔➔ $We = \left(\frac{Q*8}{100}\right)$ ➔➔ $0{,}08Q$

$\sum W = 0{,}03Q + 0{,}08Q + 469{,}48 = 13005$ ➔➔ $Q = 113959{,}27\ N$ ➔➔ $11395{,}93\ Kg$

Useful load = 11395,93 Kg – 5000 Kg = 6395,93 Kg

VEHICLE MOVEMENTS

C-)

$Q_{m}* f > Z_{tr}$ (required for starting the motion)

$Q_{m}* f = 2/3 * 113959{,}27 * 0{,}6 = 45583{,}71\ N$

$Z_{tr} = \sum W = Wy + Wh + We$

Wh=0 vehicle is not moving

$Wy = \mu_r * Q$ ➔➔ $0{,}030 * 113959{,}27$ ➔➔ $3418{,}78\ N$

$We = 0{,}08 * Q$ ➔➔ $0{,}08 * 113959{,}27$ ➔➔ $9116{,}74\ N$

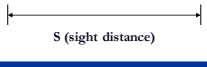
$\sum W = 12535{,}52\ N$

$45583{,}71\ N > 12535{,}52$ N so it can start moving again.

VEHICLE MOVEMENTS

SIGHT DISTANCE

Sight distance: the length of roadway visible to a driver.



S (sight distance)

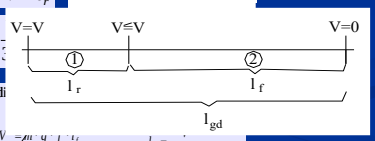
Stopping Sight Distance
(Break and safe stopping distance)

Passing Sight Distance
(Safely overtaking)

VEHICLE MOVEMENTS

Stopping Sight Distance
(Break and safe stopping distance)

$l_r = V * t_r$ t_r:Perception-reaction time



$l_r =$

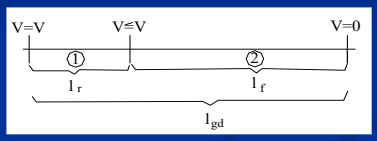
Brake di

$\frac{1}{2} * m * V^2 = m \cdot g \cdot f \cdot l_f$ $l_f = \frac{V^2}{2 * g * f}$

If the roadway is not level $\frac{1}{2} * m * V^2 = Q * f * l_f \mp Q * S * l_f$

VEHICLE MOVEMENTS

Stopping Sight Distance



Stopping time=?
Deceleration (γ) $\gamma = \frac{V - 0}{t_f}$ $t_f = \frac{V(m/sn)}{\gamma}$

$F = m * a$ $Q * f = \frac{Q}{g} * \gamma$ $f * g = \gamma$ $t_f = \frac{V}{g * f}$ $t_f = \frac{V(km/sa)}{35{,}3 * f}$

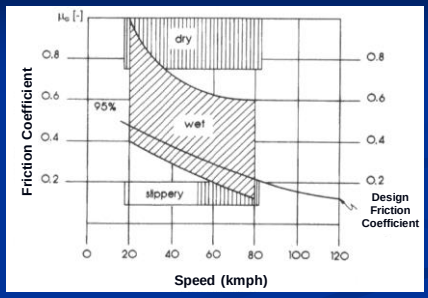
VEHICLE MOVEMENTS

Friction

Tip	Yüzey Dokusu Şekli	Açıklama
1		Mikro veya makro doku olmayan (düz)
2		Mikro doku
3		Makro (arası açılmış ve çukur) ve mikro doku
4		Düşük yoğunluklu makro dokuya sahip 3. tipte olduğu gibi, makro ve mikro doku
5		Makro doku (kaba taneler arasındaki boşluklarda) ve mikro doku (kaba tanelerin kırılmış keskin kenarları boyunca)
6		Makro doku (kaba taneler arasındaki boşluklarda)
7		Makro doku (araları açılmış ve çukur bulunuları) C

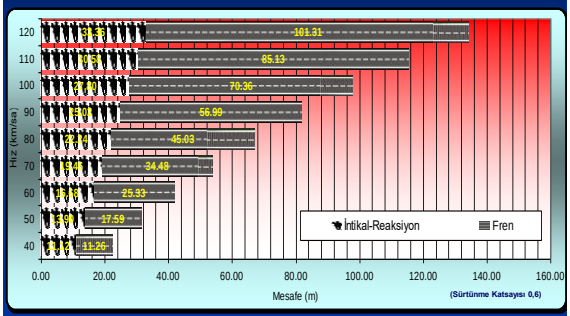
VEHICLE MOVEMENTS

Friction Coefficient



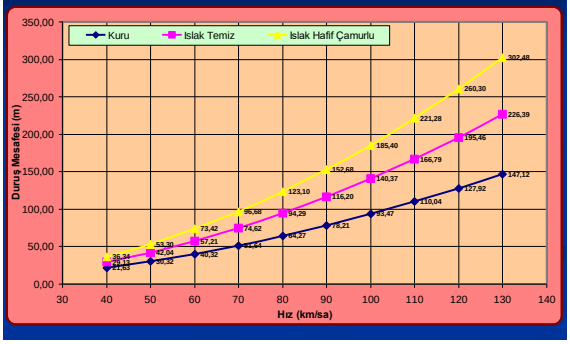
VEHICLE MOVEMENTS

Releationship between stopping distance and speed



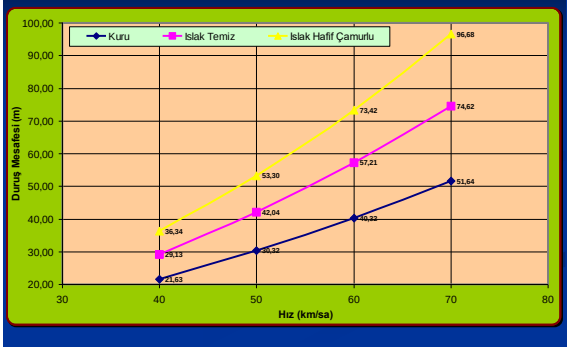
VEHICLE MOVEMENTS

Releationship between stopping distance and speed for different surface condition



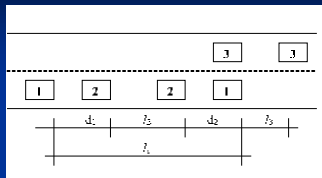
VEHICLE MOVEMENTS

Releationship between stopping distance and speed



VEHICLE MOVEMENTS

Passing Sight Distance



d : Perception-reaction time

$d_1 = a + b*V + c*V^2$

a:length of vehicle

V_1 and V_2 are constant during the overtaking.

d: following distance between vehicles

$l_2 = V_2 * t_s$ $\quad d_1 + l_2 + d_2 = l_s$

$l_s = V_1 * t_s = d_1 + V_2 * t_s + d_2$ $\quad t_s = \frac{d_1 + d_2}{V_1 - V_2}$ Passing time

$l_s = V_1 * t_s = \frac{d_1 + d_2}{V_1 - V_2} * V_1$ Passing length

VEHICLE MOVEMENTS

Passing Sight Distance

Overtake maneuver on accelerated movement



V_1, V_2 $V_1 > V_2$ Overtaken vehicle speed is constant. It is equal to V_1

Overtaking vehicle accelarat ion is γ. Initial speed of overtaking vehicle is V_1 and V_2 at the end of the overta king.

$l_2 = V_2 * t_s$ $\quad l_s = V_2 * t_r + V_2 * (t_s - t_r) + \frac{1}{2} * \gamma * (t_s - t_r)^2 = 2 * d_2 + V_2 * t_s$

$t_s = t_r + 2 * \sqrt{\frac{d_2}{\gamma}}$ $\quad l_s = V_2 * t_s + 2 * d_1 = (t_r + 2 * \sqrt{\frac{d_2}{\gamma}}) * V_2 + 2 * d_2$

If any vehicle approaches from the opposite direction:

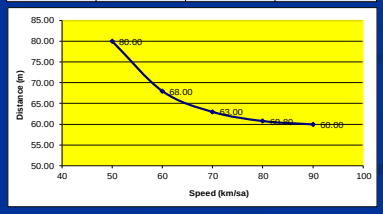
$l_g = l_s + l_3 = l_s + V_3 * t_s$

$l_{gs} = (t_r + 2 * \sqrt{\frac{d_2}{\gamma}}) * (V_2 + V_3) + 2 * d_2$

VEHICLE MOVEMENTS

Overtaking distance for various speeds

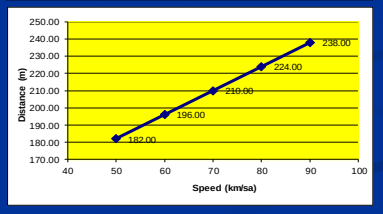
V_1 (km/sa)	V_2 (km/sa)	V_3 (km/sa)	L_s (m)
50	30	0	80.00
60	30	0	68.00
70	30	0	63.00
80	30	0	60.80
90	30	0	60.00



VEHICLE MOVEMENTS

Overtaking distance for various speeds

V_1 (km/sa)	V_2 (km/sa)	V_3 (km/sa)	L_s (m)
80	50	50	182.00
80	50	60	196.00
80	50	70	210.00
80	50	80	224.00
80	50	90	238.00



VEHICLE MOVEMENTS

Overtaking distance for various speeds

V_1 (km/sa)	V_2 (km/sa)	V_3 (km/sa)	L_s (m)
90	10	90	81.00
90	20	90	97.71
90	30	90	120.00
90	40	90	151.20
90	50	90	198.00

